Anurag Singh

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Research Interests

AI Safety and Regulations, OOD Generalisation and Uncertainty Quantification, Imprecise Probability

Education

PhD in Computer Science *April 2023-Present*
CISPA Helmholtz Center for Information Security, Germany
Advisor: Prof Krikamol Muandet

MS in Computer Science *Nov 2020- Feb 2023*
Technical University of Munich,Germany *Grade: Distinction (1.5/5)*
Advisor: Prof Debarghya Ghoshdastidar & Prof Matthias Nießner

BEng. in Information Technology *Aug 14 - May 18*
Netaji Subhas Institute of Technology, Delhi *Grade: Distinction (8.7/10)*
Advisor: Prof AV Subramanyam

Publications

1. Avoiding the Tragedy of the Commons in AI Regulation via Dynamic Licensing [pdf]
   Rajeev Verma, Anurag Singh, Christian A Naesseth, Eric Nalisnick, Krikamol Muandet
   ICLR AIMS Workshop 2026, In Submission

2. Incentive Aware AI Regulations: A Credal Characterisation [pdf]
   Anurag Singh, Julian Rodemann, Rajeev Verma, Siu Lun Chau, Krikamol Muandet
   EurIPS PAIG Workshop 2025 (Spotlight), In Submission

3. Truthful Elicitation of Imprecise Forecasts [pdf]
   Anurag Singh, Siu Lun Chau, Krikamol Muandet
   UAI 2025 (Oral)

4. Domain Generalization via Imprecise Learning [pdf]
   Anurag Singh, Siu Lun Chau, Shahine Bouabid, Krikamol Muandet
   ICML 2024 (Spotlight)

5. Robust Feature Inference: A Test-time Defense using Spectral Projections.
   Anurag Singh*, Mahalakshmi Sabanayagam*, Krikamol Muandet, Debarghya Ghoshdastidar
   TMLR 2024

6. Improving Semi-Supervised Domain Adaptation Using Effective Target Selection and Semantics [pdf]
   Anurag Singh*, Naren Doraiswamy*, Sawa Takamuku, Megh Bhalerao, Titir Dutta, Soma Biswas
   LI2D @ CVPR 2021

7. Adaptive Margin Diversity Regularizer for handling Data Imbalance in Zero-Shot SBIR. [pdf]
   Titir Dutta, Anurag Singh, Soma Biswas
   ECCV 2020 (Spotlight)

8. StyleGuide: Zero-Shot Sketch-based Image Retrieval Using Style-Guided Image Generation. [pdf]
   Titir Dutta, Anurag Singh, Soma Biswas
   IEEE Transactions on Multimedia (TMM), 2020

9. Image Corpus Representative Summarization. [pdf]
   Anurag Singh, Lakshay Virmani, AV Subramanyam
   IEEE BigMM 2019, NUS Singapore (Honourable Mention Award)

Work Experience

- **Indian Institute of Science** — Bangalore, India
  Research Associate *Advisor: Prof. Soma Biswas* — *July 2019 - Sept 2020*
  - Research on learning with limited labels and imbalanced data in different computer vision tasks like classification, domain-adaptation (DA) and Sketch-based Image Retrieval (SBIR).
  - Analyzed the generalization to target data under class-imbalance in SSDA. Designed an active learning based SSDA algorithm.
  - Analysis & Design of methods for regularization of class imbalance in Zero-shot SBIR, style-guided generalized ZS-SBIR
  - Mentored two full time junior research associates on SSDA March 2020 onwards.

- **Royal Bank of Scotland (RBS)** — New Delhi, India
  Sowtware Engineer — *June 2018 - July 2019*
  - Led access management face recognition system initiative at RBS. Conceptualized the architecture to perform recognition with low latency on edge. [MVP]
  - Reviewed vendor face recognition solutions for C-level RBS India management.
  - As a software engineering Intern in summer 2017, built a utility to replay lost brokerage messages saving $\sim$ 500 man-hours annually. Considering performance, offered a full-time position.

- **IIIT Delhi** — New Delhi, India
  Research Associate *Advisor: Prof. AV Subramanyam* — *Aug 17 - Mar 18*
  - Formulated first end-to-end deep learning model and evaluation metric for unsupervised image collection summarization. First author publication in BigMM'19 at NUS Singapore. [poster] [ppt]

Honours and Awards

- Travel Awards: UAI 2025, ISIPTA 2025, ELLIS Robust ML 2023, MLSS 2023
- Merit Scholarship for Predoctoral MSCS at UChicago (28k$ Unavailed).
- Honorable Mention Award at IEEE BigMM 2019 in NUS Singapore.
- Outstanding Research Award at NSIT in 2020 for research during undergrad.

Professional Service

- **Reviewer:** NeurIPS 2023-25, ICML 2024-26 TMLR 2024-26, AISTATS 2024-26, ICLR 2025, UAI 2026

Teaching

- **TA:** Elements of Machine Learning, Winter semester 2023, Saarland University
- **TA:** Introduction to Deep Learning, Winter semester 2021, TU Munich